

Ex. No. 4

AWS SageMaker Console Setup:

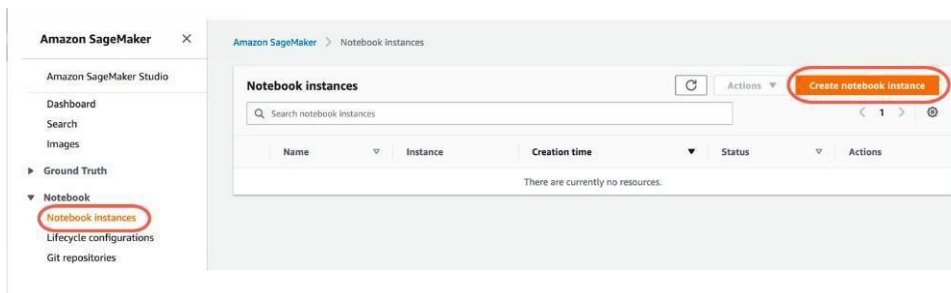
Step 1 — Create an Amazon SageMaker notebook instance

In this step, you create the notebook instance used to download and process data. As part of the creation process, you also create an IAM role that allows SageMaker to access data in Amazon S3.

a. Sign in to the Amazon SageMaker console, and in the top-right corner, select your preferred AWS Region.



b. In the left navigation pane, choose Notebook instances, then choose Create notebook instance.



c. Fill in the following fields: Notebook instance name = SageMaker-Tutorial, Notebook instance type = ml.t2.medium, Elastic inference = none.

Create notebook instance

Amazon SageMaker provides pre-built fully managed notebook instances that run Jupyter notebooks. The notebook instances include example code for common model training and hosting exercises. [Learn more](#)

Notebook instance settings

Notebook instance name

Maximum of 63 alphanumeric characters. Can include hyphens (-), but not spaces. Must be unique within your account in an AWS Region.

Notebook instance type

Elastic Inference [Learn more](#)

► Additional configuration

d. In the Permissions and encryption section, for IAM role choose Create a new role, then in the Create an IAM role dialog select Any S3 bucket and choose Create role.

Create an IAM role

Passing an IAM role gives Amazon SageMaker permission to perform actions in other AWS services on your behalf. Creating a role here will grant permissions described by the [AmazonSageMakerFullAccess](#) IAM policy to the role you create.

The IAM role you create will provide access to:

- ☒ **S3 buckets you specify - optional**
 - ☒ **Any S3 bucket**
Allow users that have access to your notebook instance access to any bucket and its contents in your account.
 - ☐ Specific S3 buckets

Comma delimited. ARNs, "" and "/" are not supported.
 - ☐ None
- ☒ Any S3 bucket with "sagemaker" in the name
- ☒ Any S3 object with "sagemaker" in the name
- ☒ Any S3 object with the tag "sagemaker" and value "true" [See Object tagging](#)
- ☒ S3 bucket with a Bucket Policy allowing access to SageMaker [See S3 bucket policies](#)

Amazon SageMaker creates the AmazonSageMaker-ExecutionRole-*** role:

Permissions and encryption

IAM role
Notebook instances require permissions to call other services including SageMaker and S3. Choose a role or let us create a role with the [AmazonSageMakerFullAccess](#) IAM policy attached.

☒ **Success! You created an IAM role.**
[AmazonSageMaker-ExecutionRole-20201204T135178](#)

Root access - optional

- ☒ **Enable** - Give users root access to the notebook
- ☐ Disable - Don't give users root access to the notebook
Lifecycle configurations always have root access

Encryption key - optional
Encrypt your notebook data. Choose an existing KMS key or enter a key's ARN.

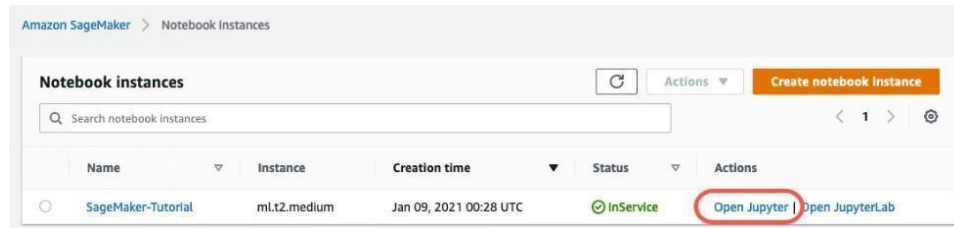
e. Keep the default settings for the remaining options and choose Create notebook instance. The new instance is shown with a status of Pending.

Amazon SageMaker > Notebook Instances

Notebook instances

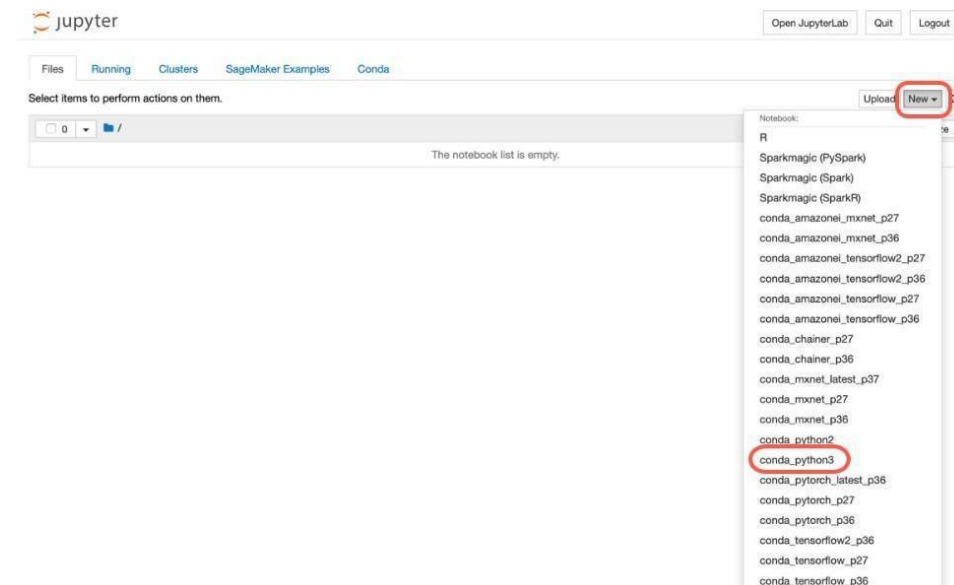
	Name	Instance	Creation time	Status	Actions
☐	SageMaker-Tutorial	ml.t2.medium	Jan 09, 2021 00:28 UTC	☒ Pending	-

The notebook is ready when the status changes to InService:

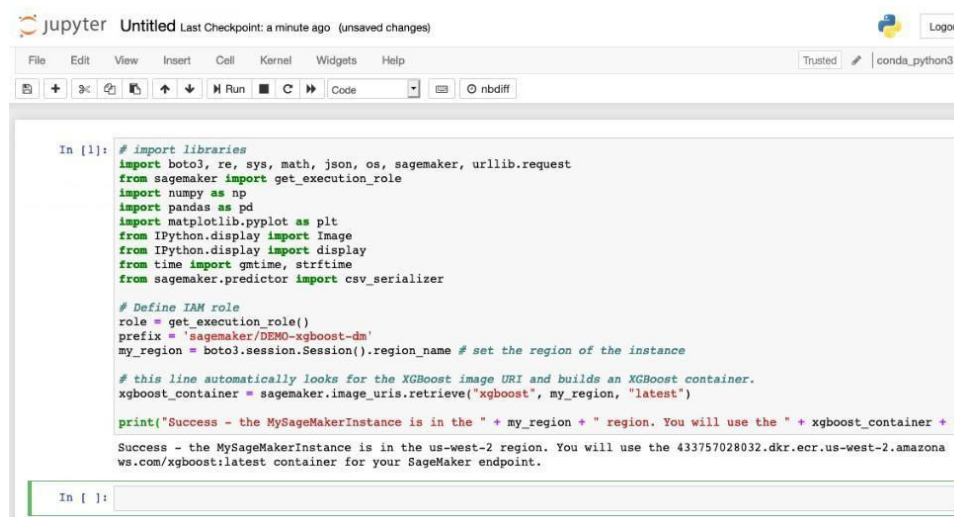


Step 2 — Prepare the data

- After the SageMaker-Tutorial notebook instance status changes to InService, choose Open Jupyter.
- In Jupyter, choose New, then choose conda_python3.



- In a new code cell, paste the import/setup code below and choose Run.



Continue running each of the remaining code sections (create bucket → download data → split data → train model → deploy → evaluate → clean up) in new cells, in order, using the code.

Program:

Import Libraries and IAM Role Setup

```
# import libraries

import boto3, re, sys, math, json, os, sagemaker, urllib.request

from sagemaker import get_execution_role

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from IPython.display import Image

from IPython.display import display

from time import gmtime, strftime


# Define IAM role

role = get_execution_role()

prefix = 'sagemaker/DEMO-xgboost-dm'

my_region = boto3.session.Session().region_name # set the region of the instance


# This automatically looks up the XGBoost image URI and builds an XGBoost container.

xgboost_container = sagemaker.image_uris.retrieve("xgboost", my_region, "latest")


print("Success - the MySageMakerInstance is in the " + my_region + " region. You will use the "

      + xgboost_container + " container for your SageMaker endpoint.")


Create the S3 Bucket (unique name, auto-generated)

# Create a globally-unique S3 bucket name instead of a hardcoded placeholder

account_id = boto3.client('sts').get_caller_identity().get('Account')

bucket_name = 'sagemaker-demo-xgboost-{}-{}'.format(account_id, strftime('%Y%m%d%H%M%S',
gmtime()))
```

```

s3 = boto3.resource('s3')

try:
    if my_region == 'us-east-1':
        s3.create_bucket(Bucket=bucket_name)
    else:
        s3.create_bucket(Bucket=bucket_name,
                          CreateBucketConfiguration={'LocationConstraint': my_region})

    print('S3 bucket created successfully: ' + bucket_name)
except Exception as e:
    print('S3 error: ', e)

```

Download and Load the Dataset

Download the data to the SageMaker instance and load it into a dataframe

```

try:
    urllib.request.urlretrieve(
        "https://d1.awsstatic.com/tmt/build-train-deploy-machine-learning-model-sagemaker/"
        "bank_clean.27f01fbbdf43271788427f3682996ae29ceca05d.csv",
        "bank_clean.csv")
    print('Success: downloaded bank_clean.csv.')
except Exception as e:
    print('Data load error: ', e)

try:
    model_data = pd.read_csv('./bank_clean.csv', index_col=0)
    print('Success: Data loaded into dataframe.')
except Exception as e:
    print('Data load error: ', e)

```

Shuffle and Split the Data

Shuffle and split the data into training data (70%) and test data (30%)

```
train_data, test_data = np.split(
    model_data.sample(frac=1, random_state=1729),
    [int(0.7 * len(model_data))])
```

```
print(train_data.shape, test_data.shape)
```

Train the XGBoost Model

Reformat the header and first column of the training data, then upload it to S3.

This is required for the Amazon SageMaker built-in XGBoost algorithm.

```
pd.concat([train_data['y_yes'], train_data.drop(['y_no', 'y_yes'], axis=1)],
          axis=1).to_csv('train.csv', index=False, header=False)
```

```
boto3.Session().resource('s3').Bucket(bucket_name).Object(
    os.path.join(prefix, 'train/train.csv')).upload_file('train.csv')
```

```
s3_input_train = sagemaker.inputs.TrainingInput(
    s3_data='s3://{}/{}/train'.format(bucket_name, prefix), content_type='csv')
```

Set up the SageMaker session, create the XGBoost estimator, and set hyperparameters

```
sess = sagemaker.Session()
xgb = sagemaker.estimator.Estimator(
    xgboost_container, role, instance_count=1,
    instance_type='ml.m4.xlarge',
    output_path='s3://{}/{}/output'.format(bucket_name, prefix),
    sagemaker_session=sess)
```

```
xgb.set_hyperparameters(  
    max_depth=5, eta=0.2, gamma=4, min_child_weight=6,  
    subsample=0.8, verbosity=0, objective='binary:logistic', num_round=100)
```

```
# Start the training job
```

```
xgb.fit({'train': s3_input_train})
```

Deploy the Model and Run Predictions

```
# Deploy the trained model to an endpoint
```

```
xgb_predictor = xgb.deploy(initial_instance_count=1, instance_type='ml.m4.xlarge')
```

```
# Predict whether customers in the test data enrolled for the bank product or not
```

```
from sagemaker.serializers import CSVSerializer
```

```
test_data_array = test_data.drop(['y_no', 'y_yes'], axis=1).values # load the data into an array
```

```
xgb_predictor.serializer = CSVSerializer() # set the serializer type
```

```
predictions = xgb_predictor.predict(test_data_array).decode('utf-8') # predict!
```

```
# Safely parse the returned CSV string of predictions into a NumPy array
```

```
# (np.fromstring is deprecated/unreliable for this; use split() + np.array instead)
```

```
predictions_array = np.array(predictions.strip().split(','), dtype='float32')
```

```
print(predictions_array.shape)
```

Evaluate Model Performance

```
# Compare actual vs. predicted values in a confusion matrix
```

```
cm = pd.crosstab(index=test_data['y_yes'], columns=np.round(predictions_array),  
                 rownames=['Observed'], colnames=['Predicted'])
```

```
tn = cm.iloc[0, 0]; fn = cm.iloc[1, 0]; tp = cm.iloc[1, 1]; fp = cm.iloc[0, 1]
```

```
p = (tp + tn) / (tp + tn + fp + fn) * 100
```

```

print("\n{0:<20}{1:<4.1f}%\n".format("Overall Classification Rate: ", p))

print("{0:<15}{1:<15}{2:>8}".format("Predicted", "No Purchase", "Purchase"))

print("Observed")

print("{0:<15}{1:<2.0f}% ({2:<}){3:>6.0f}% ({4:<})".format(
    "No Purchase", tn / (tn + fn) * 100, tn, fp / (tp + fp) * 100, fp))

print("{0:<16}{1:<1.0f}% ({2:<}){3:>7.0f}% ({4:<}) \n".format(
    "Purchase", fn / (tn + fn) * 100, fn, tp / (tp + fp) * 100, tp))

```

Clean Up Resources

Terminate the resources used in this lab to avoid ongoing charges

```
xgb_predictor.delete_endpoint(delete_endpoint_config=True)
```

```
bucket_to_delete = boto3.resource('s3').Bucket(bucket_name)
```

```
bucket_to_delete.objects.all().delete()
```

Then, from the SageMaker console:

1. Open Notebook instances

2. Select the notebook instance created for this tutorial -> Actions -> Stop

3. Once Status changes to 'Stopped' -> Actions -> Delete -> Delete